

Cyanide Poisoning

History

- Cyanide may cause toxicity through parenteral administration, inhalation, ingestion, or dermal absorption
- Have a high suspicion of cyanide exposure/toxicity in patients / firefighters with smoke inhalation from **enclosed spaces**.
- Consider in:
 - Certain industrial workers (metal trades, mining, jewelry manufacturing, and x-ray film recovery, ship fumigation)
 - Possible suicide attempt? (esp. in healthcare or lab workers)
 - Hydrogen Sulphide gas exposure (sewer gas)

Signs and Symptoms

- Presentation may vary depending on the concentration and duration of exposure
- Signs and symptoms may include, but are not limited to:
 - Nausea/Vomiting
 - Diarrhea
 - Altered mental status
 - Dyspnea
 - Abnormal skin color
 - Sweating
 - Tearing
 - Known exposure to toxic or irritating gas
 - Seizures
 - Burns to respiratory tract
 - Headache
 - Vertigo/Dizziness
 - Stridor
 - Sooty sputum
 - Constricted/dilated pupils
- **SEVERE Symptoms include:**
 - Cardiac arrest
 - Seizures
 - Cardiac Dysrhythmias
 - Respiratory difficulty / dyspnea
 - Altered mental status / coma

Notes:

- Cherry red skin is NOT seen in the majority of CN poisoned patients nor is the smell of "bitter almonds" on the breath and its absence alone should not eliminate CN toxicity as a concern
- Pulse oximetry may not be accurate for toxic inhalation victims
- Firefighter down with CPR/Seizure or pulled from Structure Fire Scene with altered mental status or signs of shock should be treated with Hydroxycobalamin (Cyanokit), when available

EMT/AEMT

- Remove patient from toxic environment
- Remove clothing and decontaminate as indicated
- Follow General Medical Care Guideline
- CO monitoring / co-oximetry / Rad-57 Monitor, if available
- High Flow **Oxygen** by Non-rebreather Mask
- **If Patient has altered mental status, signs of shock, seizures or CPR with CN exposure:**
Call for intercept with a unit that has hydroxocobalamin and immediately evacuate

Paramedic

- Consider advanced airway
- EtCO2 and CO monitoring, if available